



## MATERIAL SAFETY DATA SHEET

# Folic Acid Feed Grade

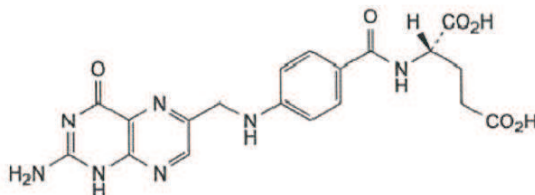
### 1. Product Identification and Use

|                     |                                                                                                                   |
|---------------------|-------------------------------------------------------------------------------------------------------------------|
| Product name        | Folic Acid Feed Grade                                                                                             |
| Trade name          | Folic Acid                                                                                                        |
| Manufacturer        |                                                                                                                   |
| Company information | JIANGXI TIANXIN PHARMACEUTICAL CO.,LTD.<br>Address: Le'anjiang Industrial Zone, Leping, Jiangxi,<br>333300, China |
| Use                 | Active pharmaceutical ingredient, nutritional ingredient.                                                         |

### 2. Composition / Information on Ingredients

|                   |                                                                                                                                                                             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chemical name     | L-Glutamic acid, N-[4-[[[(2-amino-1,4-dihydro-4-oxo-6-pteridiny]methyl]amino]benzoyl]-;<br>N-[p-[[[(2-Amino-4-hydroxy-6-pteridiny]methyl]amino]benzo<br>yl]-L-glutamic acid |
| CAS No.           | 59-30-3                                                                                                                                                                     |
| EINECS No.        | 200-419-0                                                                                                                                                                   |
| Molecular formula | C <sub>19</sub> H <sub>19</sub> N <sub>7</sub> O <sub>6</sub>                                                                                                               |
| Molecular weight  | 441.40 g/mol                                                                                                                                                                |

Constitution



### 3. Hazards Identification

|            |                                                                                                                               |
|------------|-------------------------------------------------------------------------------------------------------------------------------|
| Eyes       | May cause eye irritation. The toxicological properties of this material have not been fully investigated.                     |
| Skin       | May cause skin irritation. The toxicological properties of this material have not been fully investigated                     |
| Ingestion  | May cause irritation of the digestive tract. The toxicological properties of this substance have not been fully investigated. |
| Inhalation | May cause respiratory tract irritation. The toxicological properties of this substance have not been fully investigated.      |

### 4. First Aid Measures

|                      |                                                                                                                                                                                                                                    |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Overexposure to Eyes | Immediately flush thoroughly with water for at least 15 minutes, occasionally lifting the upper and lower eyelids.<br>Get medical aid immediately.                                                                                 |
| Skin                 | Flush skin with plenty of soap and water for at least 15 minutes while removing contaminated clothing and shoes.<br>Get medical aid if irritation develops or persists.<br>Wash clothing before reuse.                             |
| In case of Ingestion | Never give anything by mouth to an unconscious person.<br>Get medical aid. Do NOT induce vomiting. If conscious and alert, rinse mouth and drink 2-4 cupfuls of milk or water.<br>Get medical aid if irritation or symptoms occur. |
| Inhalation           | Remove from exposure to fresh air immediately.<br>If not breathing, give artificial respiration. If breathing is difficult, give oxygen.<br>Get medical aid if cough or other symptoms appear.                                     |
| Notes to physician   | Treat symptomatically and supportively.                                                                                                                                                                                            |

### 5. Fire Fighting Measures

|                            |               |                                                                                                                        |
|----------------------------|---------------|------------------------------------------------------------------------------------------------------------------------|
| Suitable media             | extinguishing | Use water spray, dry chemical, carbon dioxide, or appropriate foam.                                                    |
| Special procedures         | firefighting  | Wear self-contained breathing apparatus pressure-demand and protective clothing to prevent contact with skin and eyes. |
| Unusual explosions hazards | fire and      | Emits toxic fumes under fire conditions.                                                                               |

## 6. Accidental Release Measures

|                |                                                                                                                                                                                                            |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Spill response | Wear respirator, chemical safety goggles, rubber boots and heavy rubber gloves.<br>Sweep up, then place into a suitable container for disposal.<br>Avoid generating dusty conditions. Provide ventilation. |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 7. Handling and Storage

|          |                                                                                                                                                                                                                                                                                          |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Handling | Wash thoroughly after handling, remove contaminated clothing and wash before reuse.<br>Use with adequate ventilation.<br>Minimize dust generation and accumulation.<br>Avoid contact with eyes, skin, and clothing.<br>Avoid ingestion and inhalation.<br>Keep container tightly closed. |
| Storage  | Preserve in well-closed, light-resistant containers. Protected from light.                                                                                                                                                                                                               |

## 8. Exposure Controls / Personal Protection

|                     |                                                                                                                                                                             |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Personal Protection |                                                                                                                                                                             |
| Eyes                | Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166. |
| Skin                | Wear appropriate protective gloves to prevent skin                                                                                                                          |



exposure.

#### Clothing

Wear appropriate protective clothing to prevent skin exposure.

#### Respirators

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Always use a NIOSH or European Standard EN 149 approved respirator when necessary.

### 9. Physical and Chemical Properties

Color Yellow-orange

Physical state Powder solid

Odor Odorless

#### Solubility

It readily dissolves in dilute solutions of alkali hydroxides and carbonates. Soluble in hot, 3 N hydrochloric acid, in hot, 2 N sulfuric acid, in hydrochloric acid, and in sulfuric acid; very slightly soluble in water; insoluble in alcohol, in acetone, in chloroform, and in ether.

Melting point 250°C (with decomposition)

### 10. Stability and Reactivity

Chemical stability Stable under normal temperatures and pressures.

Material to avoid Strong oxidizing agents

Conditions to avoid Incompatible materials, light, dust generation.

Hazardous decomposition products Carbon monoxide, oxides of nitrogen, irritating and toxic fumes and gases, carbon dioxide.

### 11. Toxicological Information

Toxicity data LD<sub>50</sub> 10 gm/kg (oral, mouse)

Acute effects May cause skin and eye irritation or corrosion.

Carcinogenicity No classification data on carcinogenic properties of this

material id available from the EPA, IARC, NTP, OSHA or ACGIH.

## 12. Ecological Information

### Toxicity

Aquatic toxicity: No further relevant information available

Persistence and degradability No further relevant information available

Bioaccumulative potential No further relevant information available

Mobility in soil No further relevant information available

### Additional ecological information:

Do not allow product to reach ground water, water bodies or sewage system.

Do not allow material to be released to the environment without proper governmental permits.

General notes Water hazard class 1 (Self-assessment): slightly hazardous for water.

Do not allow undiluted product or large quantities of it to reach ground water, water bodies or sewage syystem.

Avoid transfer into the environment.

### Results of PBT and vPvB assessment

PBT Not applicable

vPvB Not applicable

Other adverse effects No further relevant information available

## 13. Disposal Considerations

EPA Waste Number Not available.

|                     |                                                                                                                                                                                      |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Treatment           | Hand over to disposers of hazardous waste.<br>Must be specially treated under adherence to official regulation.<br>Consult state, local or national regulations for proper disposal. |
| Waste from residues | Observe all national and local environmental regulations.                                                                                                                            |

#### 14. Transport Information

|                          |                                                   |
|--------------------------|---------------------------------------------------|
| DOT Classification       | Proper Shipping Name: None<br>RQ: Not applicable. |
| TDG Classification       | Not applicable.                                   |
| IMO/IMDG Classification  | No classified as hazardous.                       |
| ICAO/IATA Classification | Not applicable.                                   |

#### 15. Regulatory Information

|                                                            |                                                                                                                                                              |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ELINCS (European List of Notified Chemical Substances)     | Substance is not listed                                                                                                                                      |
| Australian inventory of Chemical substance                 | Substance is listed                                                                                                                                          |
| Standard for the Uniform Scheduling of Drugs and Poisons   | Substance is not listed                                                                                                                                      |
| Information about limitation of use                        | For use only by technically qualified individuals                                                                                                            |
| Water hazard class                                         | Water hazard class 1 (Self-assessment): slightly hazardous for water.                                                                                        |
| Other regulations, limitations and prohibitive regulations | Although this chemical may sometimes be used as a food or drug or cosmetic, our products are not approved or suitable for such use or for human consumption. |



Substances of very high concern (SVHC) according to REACH, Article 57

Substance is not listed

REACH-Pre-registered substances

Substance is not listed

Chemical assessment

safety A chemical Safety Assessment has not been carried out.

## 16. Other Information

Use

Active pharmaceutical ingredient, nutritional ingredient, feed additive

Last update: Jan 1, 2018

The information in this MSDS is based on current scientific knowledge. It should not be taken as expressing or implying any warranty concerning product characteristics.

Shengyan Li QA Manager

